Zirui Ren (任梓瑞)

Third-year Undergraduate, Department of Physics, Tsinghua University, Beijing, China | Date of Birth: April 2006
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EDUCATION

Tsinghua University, Beijing, China — Sep. 2023 | Present
Major in Physics | Minor in Computer Science and Technology — Cumulative GPA: **3.93 / 4.0**

- **Selected Awards:** Tsinghua Comprehensive Excellence Scholarship (Top 5%, 2024 & 2025).
- **Language Proficiency:** TOEFL 113/120 | CET-6 665/710

PUBLICATIONS

- **Are Your Reasoning Models Reasoning or Guessing? A Mechanistic Analysis of Hierarchical Reasoning Models**
  Zirui Ren, Ziming Liu
  arXiv:2601.10679. *Under review at **NeurIPS 2026** (First Author)*
- **LLM × MapReduce-V2: Entropy-Driven Convolutional Test-Time Scaling for Generating Long-Form Articles from Extremely Long Resources**
  Haoyu Wang, Yujia Fu, Zhu Zhang, Shuo Wang, **Zirui Ren**, Xiaorong Wang, Zhili Li, Chaoqun He, Bo An, Zhiyuan Liu, Maosong Sun
  arXiv:2504.05732

RESEARCH & INTERNSHIP EXPERIENCE

AI for AI: Meta-Modeling for Pre-training Dynamics & Hyperparameter Extrapolation — Jan. 2026 | Present
Advisor: Prof. Ziming Liu, College of AI, Tsinghua University

- **Meta-Modeling beyond Scaling Laws:** Pioneering an "AI for AI" paradigm to break the compute bottleneck of LLM pre-training. Developing a neural meta-model trained to accurately predict the training trajectories and performance of target foundation models across various architectures and hyperparameter spaces.
- **Inverse Optimization:** Engineering a differentiable pipeline that leverages the meta-model's predictions to deduce optimal pre-training configurations. This approach aims to extrapolate to large-scale models in dimensions beyond traditional scaling law.

Mechanistic Analysis of Reasoning Models — Jul. 2025 | Mar. 2026
Collaboration with MIT Advisor: Dr. Ziming Liu

- **Discovery of Latent Anomalies:** Conducted a systematic mechanistic study on Hierarchical Reasoning Models (HRMs). By probing latent-state trajectories and structural geometry, identified critical violations of fixed-point assumptions, proving that advanced reasoners exhibit a hidden "guessing" shortcut instead of robust rule-based deduction.
- **Intervention & Optimization:** Engineered three training-free representation-steering and inference-guidance techniques derived from mechanistic insights, drastically boosting HRM accuracy on *Sudoku-Extreme* from 55% to **97%**.
- **Infrastructure:** Independently implemented the end-to-end distributed evaluation and probing pipeline (PyTorch), maintaining the open-source codebase. (Resulted in a first-author paper).

Research Intern (Foundation Model Architecture & Distillation) — Jun. 2025 | Aug. 2025
ModelBest Inc. (面壁智能) Advisor: Prof. Zhiyuan Liu, Tsinghua University

- **Architecture Exploration for SALA-9B:** Served as an early-stage collaborator exploring sub-quadratic model architectures. Conducted fine-grained ablation studies on hybrid lightning-attention modules in 0.6B-scale models.
- **Pre-training & Evaluation:** Executed a 3-stage progressive knowledge distillation pipeline from Qwen3-0.6B via the RAD-LADS protocol. Tracked training stability and evaluating benchmarks (MMLU, HellaSwag, PIQA, etc.), laying the empirical foundation for ModelBest's million-context **SALA-9B** model.

Research Intern (Test-Time Scaling for Long-to-long Generation) — Feb. 2025 | Aug. 2025
ModelBest Inc. (面壁智能) Advisor: Prof. Zhiyuan Liu, Tsinghua University

- **Algorithm Design:** Co-developed `LLM × MapReduce-V2`, a novel test-time scaling framework borrowing CNN convolutional concepts to map ultra-long contexts into structured, multi-stage reasoning graphs for high-fidelity long-form text generation.
- **Implementation:** Successfully scaled the algorithmic prototype into **SurveyGO**, ModelBest's production-level academic survey platform. Designed the engine handling incremental text refinement conditioned on dynamic streaming references.

SELECTED COURSES

- Deep Reinforcement Learning (Ongoing)
- Optimization Theory (A)
- Introduction to High Performance Computing (Ongoing)
- Computer Organization and Architecture (Pass)
- Data Structures (A)
- Linear Algebra (A+)
- Quantum Mechanics (A+)
- Linear Regression Analysis (A)
- Numerical Analysis (A-)
- Topology (A)

TECHNICAL SKILLS

- **Coding:** Python, PyTorch, C/C++, CUDA/Triton.
- **Domain Expertise:** Large Language Model Pre-training / Knowledge Distillation, Linear Attention Mechanisms, Test-time Scaling, Mechanistic Interpretability.